WYI System Manual

Intelligent Bill Management and Review Platform
F# Backend + Vue.js Frontend + Gemini AI

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Abstract

The WYI project is a multi-modal AI-driven platform [cite: 3] designed to automate the recognition, classification, and validation of financial documents[cite: 3]. This manual serves as a comprehensive guide to the system architecture, component management, and AI integration protocols[cite: 4].

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Chapter 1

Project Introduction

1.1 Objective

The primary goal of WYI is to minimize manual data entry in bill management[cite: 6]. By utilizing the Google Gemini 1.5 Flash model, the system achieves high-accuracy parsing of complex document layouts[cite: 6].

1.2 System Requirements

- **Runtime:** .NET 9.0 Core with F# [cite: 7]
- **Database:** SQL Server / PostgreSQL [cite: 8]
- **Authentication:** Clerk.js Integration [cite: 12]
- **AI Engine:** Google Gemini Multimodal API [cite: 6, 10]

Chapter 2

System Features

2.1 Provider Management

The system maintains a comprehensive database of service providers. Users can manage metadata, account numbers, and historical billing patterns for each provider to improve future recognition accuracy.

Internet | Spectrum | Comcast | Windstream
Cellphone | Verizon | AT&T | T-Mobile
Landlines
Cable TV | Optimum
Satellite TV | Direct TV | Dish
Satellite Radio | Sirius XM
Security Monitoring | ADT | Vivint | Brinks
Payroll | ADP | Paychex
Credit Card Processing | Global | World Pay
Trash | Waste Mgmt. | Republic | GFL | Local companies
Pest Control | Terminix | Orkin
Lawn Care | TruGreen
Subscriptions/Memberships | Adobe | Salesforce | HubSpot
Advertising | Billboards | Newspapers | Magazines | Yellow Pages
Water Delivery | Ready Refresh | Primo | Crystal Rock
Propane Gas
Heating Oil
Elevator Maintenance Contracts | Otis
Monthly Recurring Bills

Figure 2.1: Provider Management Interface: Managing service entities and account details.

2.2 AI-Powered Auto-Filling

The core innovation of WYI is the automated extraction of fields such as Amount, Bill Date, and Category from uploaded images or PDFs.

Provider:

CenturyLink

Account Number:

6514645851088

Account Name:

BREW FITNESS

Address:

808 LAKE ST S

Town:

FOREST LAKE

State:

MN

ZIP:

55025-2614

Figure 2.2: AI Analysis View: Automatic field extraction and form pre-filling using Gemini 1.5 Flash.

Chapter 3

System Architecture

3.1 Backend: UtilKestrel

The backend handles critical logic including file orchestration, secure data persistence, and AI prompt engineering[cite: 7, 8]. It leverages F# for robust asynchronous processing[cite: 7, 10].

3.2 Frontend: Vue 3 Ecosystem

The frontend manages user interactions and multi-file uploads via the `UploadBills` module[cite: 9]. State is managed through a reactive store to ensure consistency between the AI analysis and the UI[cite: 9].

Chapter 4

Technical Implementation

4.1 Gemini Multimodal Analysis

The following implementation demonstrates the asynchronous task for processing document images through the Gemini API[cite: 10, 11].

```
// GeminiMultimodal Task Implementation
let GeminiMultimodal output apiKey model prompt files = task {
    use client = new HttpClient()
    // Extended timeout to handle multi-image processing
    client.Timeout <- TimeSpan.FromMinutes(5.0)

    // API endpoint construction and POST execution
    let! response = client.PostAsync(url, content)
    return! response.Content.ReadAsStringAsync()
}
```

Listing 4.1: F# Gemini Integration Task

Chapter 5

Troubleshooting

5.1 Network and Authentication

If a `NetworkError` occurs during Clerk.js authentication, ensure `clerk.accounts.dev` is accessible[cite: 12]. In restricted regions, using a VPN with the Lightway protocol (UDP/TCP) is recommended to maintain a stable session[cite: 13].

5.2 Timeout Issues

A `TaskCanceledException` typically indicates that the AI analysis exceeded the default 100-second window[cite: 14]. The system is configured to use a 300-second timeout to accommodate high-resolution uploads[cite: 14].